

|   | Component's name       | Purpose                          | Location                                   |
|---|------------------------|----------------------------------|--------------------------------------------|
| 1 | Label                  | To display a label               | Palette-->User Interface-->Label           |
| 2 | Button                 | To trigger events                | Palette-->User Interface-->Button          |
| 3 | Horizontal arrangement | To arrange the child components  | Palette-->Layout-->Horizontal arrangement  |
| 4 | Notifier               | To display on-screen information | Palette-->User Interface-->Notifier        |
| 5 | Canvas                 | To enable drawing                | Palette--> Drawing and Animation--> Canvas |
| 6 | Firebase DB            | To store data persistently       | Palette-->Experimental-->Firebase DB       |

Table 1: Components required to build the XDXD game app

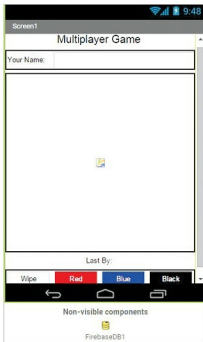


Figure 1: Designer screen



Figure 2: Application outlook

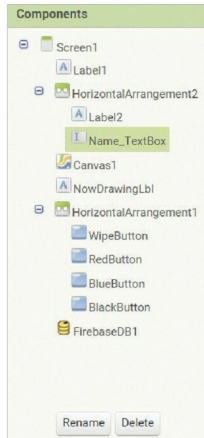


Figure 3: Components view

are special components that keep all child components horizontally aligned.

4. **Notifier:** This is used to display some instructions or give controls over your existing components. You will see its functionality in more detail as we implement it in our game.
5. **Canvas:** This is the component for drawing and for animation. You can place various animation objects over the canvas to control them via user actions. We will see more details as we go along.

6. **Firebase DB:** As you already know, this is the cloud based database utility from Google. We will be using it to store the user's data, over the cloud.

The components that we will require for this application are given in Table 1.

1. We will drag them from the left hand side palette on to the designer.
1. Drag and drop the components mentioned in Table 1 to the viewer.
2. Visible components can be seen by you, while the non-visible components will be located beneath the viewer under the tag 'Non-visible'.
3. We have placed a label to put the name of the application.
4. All buttons need to be put within the horizontal arrangement to keep them aligned horizontally.
5. If you have dragged and placed everything, the layout will look something like what's shown in Figure 1.
6. Make the necessary property changes like we did when changing the text property for the label and button components. Table 2 depicts this more clearly.

Renaming the components helps to identify them in the block editor.

Your graphical user interface is now ready. The application will look exactly like what's shown in Figure 2, after the installation.

Figure 3 shows the hierarchy of the components that we have dragged to the designer.

If you are confused seeing the designer and the components' viewer, let me explain things a bit. Here is the hierarchy that we have placed for our application.